

Relevant Knowledge/Training Skills

I have direct research experience in the field and therefore I possess several relevant skills to tackle a PhD project in the domain: programming, simulation, and hands-on lab skills.

I have ample programming experience. Particularly, I am proficient in Python as I used the language to design scripts for physics research. For example, in my work for Low Gain Avalanche Detectors (LGAD), I created a script to calculate the time resolution of LGAD from a dedicated experiment using Sr-90. Additionally, I designed a Python script used for measurement automation of an accelerator mass spectrometer (AMS) at HZDR. I additionally have programming experience outside Python. Specifically, I have used C++/ROOT to analyse test beam data of ATLAS ITk strip detectors and contributed to the development of the simulation framework for silicon detectors called Allpix².

Regarding simulation, I implemented a new hexagonal pixel model within the Allpix2 simulation framework during my CERN work. The model was then used to simulate basic performance tests of hexagonal pixels such as cluster size and hit map.

Lastly, I have extensive hands-on lab experience. For example, in my LGAD research work, I have extensively worked in my university's clean room building a dedicated Sr-90 experiment for measuring the time resolution of LGAD. Additionally, I was also involved assembling a test beam-line prototype of an AMS.

Please give a brief indication of your likely topic

The topic will be the development of a novel semiconductor sensor called Low Gain Avalanche Detector (LGAD) for potential applications involving particle physics, synchrotrons, and medical fields. Current semiconductor devices reach a spatial resolution in the order of 20 μm and a temporal resolution in the order of 1 ns. However, LGAD shows the promise of providing a high spatial and temporal resolution. Particularly, recent innovative LGAD designs such as iLGAD and trLGAD show promise in achieving a 20-micrometre spatial resolution while achieving a temporal resolution in the order of 20 ps. The research project will be about exploring and characterising the different design choices and parameters of LGAD in order to reach a very high spatial and temporal resolution.

Please give a short statement that describes your academic interests, purpose, objectives and motivation in undertaking this postgraduate study.

My enthusiasm for experimental particle physics stems from my involvement in research outside my degree for CERN's upcoming HL-LHC upgrade. This first started in 2019 when I took on a research project within my university's particle physics experiment group, where the project focused on performing timing resolution measurements of Low Gain Avalanche Detectors (LGAD). As I enjoyed research on the instrumentation aspect of particle physics, I took a role at Rutherford-Appleton Laboratory (RAL) to analyse test beam data of ATLAS ITk Strip detectors and finally at CERN I contributed to the simulation framework for semiconductor sensors called Allpix-squared. During these projects, I particularly enjoyed the research theme on the HL-LHC upgrade. Therefore, I have taken up research outside my degree and joined my university's LGAD group involving institutions and researchers from Edinburgh, Manchester, CERN, AGH Kraków, Uppsala, and IMB-CNM. It was during this involvement that I learned more in-depth how LGAD's fast timing could benefit Particle physics experiments and subsequently gave me the opportunity to present my research at CERN's

39th RD50 Workshop and the LHCb-UK's 2022 Annual Meeting. Involvement in the group further taught me how research requires adaptability, persistence, and tenacity to push it forward. Therefore, by taking on a PhD project at the University of Edinburgh, I want to challenge myself to the limit, improve my research skills, and make a further meaningful contribution to the field.

